

Ava Yunus

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SUMMARY

Third-year Software Engineering co-op student and Robotics Intern at the Canadian Space Agency, supporting robotics development for the Canadian Lunar Rover mission. Over one year of hands-on experience building full-stack and embedded systems, with experience in Python, TypeScript, Java, backend development, and API design.

EDUCATION

York University, Lassonde School of Engineering | Toronto, ON

Spec. Hons. Bachelor of Engineering, Software Engineering (Co-op) | Expected 2028

Relevant Coursework: Operating Systems, Computer Organization, Data Structures & Algorithms, Embedded Systems

EXPERIENCE

Robotics Intern | Canadian Space Agency | Fall 2026 - Present

Space Exploration Directorate | Longueuil, QC

- Support internal robotics projects involving rovers and sensors such as lidar and cameras for the Canadian Lunar Rover mission.
- Support development of a map-based mission planning tool to evaluate rover trajectories against lighting, communication access, power, and operational constraints.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C++, SQL, HTML/CSS, Bash

Backend: FastAPI, Node.js, REST APIs, SQLite, PostgreSQL

Frontend: Next.js, React, Tailwind CSS, Leaflet | **Tools:** Git, Docker, Linux, Vercel, Figma

PROJECTS

Vigil | Next.js, FastAPI, Python, Leaflet, SQLite | 2025

- Built a real-time event monitoring dashboard aggregating two external sources, with categorized events on an interactive map and severity-based filtering.
- Developed a FastAPI backend with SQLite storage and API endpoints for event ingestion and retrieval.

RingLight | C++, ESP32, WS2812B | 2025 - Present

- Designing a custom ring-shaped desk lamp with individually addressable LEDs, collaborating with a senior engineer through design reviews and iterative prototyping.
- Writing ESP32 firmware in C++ for WS2812B LED animations, managing timing constraints and memory limitations.

Lecture Transcription Plugin for Obsidian | TypeScript, esbuild, Obsidian API | 2026 - Present

- Developing a plugin that transcribes live lectures into structured Markdown notes for searchable, linked lecture archives.
- Building in TypeScript on Linux with esbuild, integrating with Obsidian APIs for editor and vault interactions.

Flexx Fitness Application | Java, SQL | 2022

- Implemented authentication, user flow, and navigation for a team-built desktop application; refined onboarding through user testing and UX iteration.

LEADERSHIP

Robotics Mentor & Project Lead | John Paul II Catholic Secondary School | 2019 - 2020

- Mentored students in prototyping and debugging for robotics competitions; supported team problem-solving through structured documentation and iterative development.

ACHIEVEMENTS

UNHack Competition, BEST Lassonde | 3rd Place Winner

Avie Bennett First Generation Award for Continuing Students | 2021

BEST Technology Impact Prize | 2021

Bank of Montreal Award | 2021